

Method Development Plan

Synthetic example for a non-existing validation method that must be developed before execution.

Development charter

Development ID	DEV-005
Proposed method	Pressure Pulse Endurance
Programme	VP-ECHO / Project Orion
Owner	Test Development Engineering
Target release	2026-09-18
Reason	No released method currently meets pressure amplitude / waveform requirement

Work breakdown

Stage	Owner	Effort	Exit criterion
Requirement clarification	Test Engineer	8 h	Acceptance limits and waveform agreed
Concept / method design	Development Engineer	16 h	Concept review complete
Fixture design	Mechanical Engineer	24 h	Drawing + safety review
Software / script	Automation Engineer	16 h	Versioned script passes bench test
Feasibility trial	Technician + Engineer	1 day	Stable pressure waveform demonstrated
Correlation / validation	Development Engineer	2 days	Repeatability and reference correlation accepted
Documentation / release	Reviewer	8 h	Controlled method revision released

Development risk & lessons

Historical signal	Preventive action
Connector leakage seen in prior pulse rigs	Pressure-decay precheck before every trial
Specification ambiguity on peak definition	Agree peak/overshoot definition before fixture freeze
DAQ aliasing on fast transients	Confirm acquisition bandwidth in feasibility gate
Single qualified specialist risk	Train a second independent operator before release

Cost estimate

Engineering	Technician	Fixture	External	Overhead / contingency	Total
EUR 6,200	EUR 1,450	EUR 2,500	EUR 500	EUR 1,930	EUR 12,580